

WEEK – 4th TASK

LIBRARY MANAGEMENT SYSTEM

Topic: Database Design, Book Issue/Return, Updates and Member Borrowing History

Objective

Design and implement a Library Management Database using SQL. The system manages members, books, book categories, borrowing transactions, issue/return dates, fines, and member borrowing history.

Requirements

1. Create database and tables.
2. Store member and book information.
3. Manage book issue and return transactions.
4. Store issue date, return date, quantity/status and fine amount.
5. Perform insertion and modification operations.
6. Generate member borrowing history reports.

SQL PROGRAM

```
CREATE DATABASE LibraryManagementDB;
USE LibraryManagementDB;

-- =====
-- 1. MEMBER TABLE
-- =====
CREATE TABLE Member (
    member_id INT PRIMARY KEY,
    member_name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    phone VARCHAR(15)
);

-- =====
-- 2. BOOK TABLE
-- =====
CREATE TABLE Book (
    book_id INT PRIMARY KEY,
    book_name VARCHAR(150) NOT NULL,
    author VARCHAR(100),
    category VARCHAR(50)
);

-- =====
-- 3. BORROW TABLE
-- =====
CREATE TABLE Borrow (
    borrow_id INT PRIMARY KEY,
    member_id INT,
    book_id INT,
    issue_date DATE,
    return_date DATE,
    status VARCHAR(20),
    fine_amount DECIMAL(10,2),
    FOREIGN KEY (member_id) REFERENCES Member(member_id),
    FOREIGN KEY (book_id) REFERENCES Book(book_id)
);

-- =====
-- 4. INSERT MEMBERS
```

```

-- =====
INSERT INTO Member
(member_id, member_name, email, phone)
VALUES
(1, 'Arun Kumar', 'arun@gmail.com', '9876543210'),
(2, 'Priya Sharma', 'priya@gmail.com', '9876543211'),
(3, 'Karthik Raj', 'karthik@gmail.com', '9876543212'),
(4, 'Sneha Devi', 'sneha@gmail.com', '9876543213'),
(5, 'Rahul Das', 'rahul@gmail.com', '9876543214');

-- =====
-- 5. INSERT BOOKS
-- =====
INSERT INTO Book
(book_id, book_name, author, category)
VALUES
(101, 'Python Programming', 'Mark Lutz', 'Programming'),
(102, 'Database System Concepts', 'Abraham Silberschatz', 'Database'),
(103, 'Artificial Intelligence', 'Stuart Russell', 'AI'),
(104, 'Web Development Basics', 'Jon Duckett', 'Web'),
(105, 'Data Structures', 'Seymour Lipschutz', 'Computer Science'),
(106, 'Machine Learning', 'Tom Mitchell', 'AI');

-- =====
-- 6. INSERT BORROW RECORDS
-- =====
INSERT INTO Borrow
(borrow_id, member_id, book_id, issue_date, return_date, status, fine_amount)
VALUES
(1001, 1, 101, '2026-08-10', '2026-08-15', 'Returned', 0.00),
(1002, 2, 102, '2026-08-11', NULL, 'Issued', 0.00),
(1003, 1, 105, '2026-08-12', '2026-08-18', 'Returned', 20.00),
(1004, 3, 106, '2026-08-13', NULL, 'Issued', 0.00),
(1005, 4, 103, '2026-08-14', '2026-08-20', 'Returned', 0.00);

-- =====
-- 7. DISPLAY MEMBERS
-- =====
SELECT * FROM Member;

-- =====
-- 8. DISPLAY BOOKS
-- =====
SELECT * FROM Book;

-- =====
-- 9. DISPLAY BORROW RECORDS
-- =====
SELECT * FROM Borrow;

-- =====
-- 10. UPDATE BORROW RECORD
-- =====
UPDATE Borrow
SET status = 'Returned',
    return_date = '2026-08-25'
WHERE borrow_id = 1002;

-- =====
-- 11. UPDATE FINE
-- =====
UPDATE Borrow
SET fine_amount = 30.00
WHERE borrow_id = 1004;

-- =====
-- 12. MEMBER BORROWING HISTORY
-- =====
SELECT
    Member.member_name,
    Borrow.borrow_id,
    Book.book_name,
    Book.author,
    Borrow.issue_date,

```

```
Borrow.return_date,  
Borrow.status,  
Borrow.fine_amount
```

```
FROM Member
```

```
JOIN Borrow  
  ON Member.member_id = Borrow.member_id
```

```
JOIN Book  
  ON Borrow.book_id = Book.book_id
```

```
ORDER BY Borrow.issue_date;
```

Expected Output / Result

The Library Management System successfully creates and manages member, book, and borrowing information. The SQL operations allow records to be inserted, updated, displayed, and combined using JOIN queries to generate a complete member borrowing history.

Conclusion

This assignment demonstrates the use of relational database concepts, primary keys, foreign keys, INSERT, SELECT, UPDATE, and JOIN operations. The system can be extended with additional features such as book availability, reservation management, and automated fine calculation.